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2027 CYCLE

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CUET UG

Chemistry

Coordination Compounds

Official Syllabus Unit V · Class 12 · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus 2026, the current 2027 baseline, Unit V)	Skills the questions test
Definitions and Werner's theory	General Introduction to Coordination/ complex compounds, Werner's theory, double salts, Basic definitions: coordination entity, central atom/ion, Ligands, coordination number, coordination sphere, coordination polyhedron, oxidation number of central atom, homoleptic and heteroleptic complexes	Labelling every part of a complex formula, computing oxidation number and coordination number (bidentates count twice), and contrasting complexes with double salts.
IUPAC nomenclature	IUPAC nomenclature of coordination compounds	Writing systematic names from formulas and formulas from names, with ligand alphabetisation and oxidation states in brackets.
Isomerism	Isomerism in coordination compounds	Classifying ionisation, hydrate, linkage and coordination isomerism, and predicting cis/trans and optical cases from the formula shape.
Bonding: VBT and CFT	Bonding in coordination compounds- Valence Bond theory (VBT) and its limitations, crystal field theory (CFT): applications (magnetic properties and colour of complex compounds) and limitations, spectrochemical series, Shape and Bonding in metal carbonyls	Matching hybridisation to magnetism in VBT, using the spectrochemical series to predict high/low spin, colour and moments in CFT, and explaining the synergic bond in carbonyls.
Applications	Importance & Applications of coordination compounds	Citing cyanide extraction, EDTA titrations, cis-platin and the metal centres of haemoglobin, chlorophyll and vitamin B^{12}.

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EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Chemistry paper maps strictly to NCERT Class 12, and Unit V blends definition-checking, naming drills and CFT predictions — a unit where careful reading beats raw calculation.

The Vocabulary That Runs This Whole Unit

Mix silver nitrate with ammonia and something quietly remarkable happens: the silver ion, which normally precipitates as chloride with salt water, now stays dissolved — it has been captured by ammonia molecules holding it with their lone pairs. A **coordination compound** (or complex) is a metal centre bonded to a ring of surrounding ions or molecules — its **ligands** — through **coordinate bonds**: shared electron pairs where **both** electrons come from the ligand. Alfred Werner founded this whole subject around 1898 by studying cobalt ammines such as

$$CoCl_3 \cdot 6NH_3$$

.

- **Werner's insight:** every cobalt(III) complex carried a fixed "inner sphere" of ammonia molecules that survived reactions untouched, while only "outer" chlorides reacted with silver nitrate. For

  $$CoCl_3 \cdot 6NH_3$$

  , three moles of AgCl precipitate; for

  $$CoCl_3 \cdot 5NH_3$$

  , only two — the missing chloride has moved inside. Primary valence (oxidation state, ionisable) vs secondary valence (coordination number, fixed geometry) — that two-valence picture won Werner the 1913 Nobel Prize.
- **Complex vs double salt:** dissolve $KCl \cdot MgCl_2 \cdot 6H_2O$ (carnallite) and you detect every ion individually — it keeps its identity only in the crystal. Dissolve

  $$K_4[Fe(CN)_6]$$

  and no free

  $$Fe^{2+}$$

  test works — the complex ion $[Fe(CN)_6]^{4-}$ survives intact in solution. That solution-test difference is the standard one-line discriminator.

The parts of a complex, named

Take

$$K_4[Fe(CN)_6]$$

apart — every word of the vocabulary applies at once:

- **Coordination entity:** the square-bracketed $[Fe(CN)_6]^{4-}$ — central atom plus ligands, behaving as one unit (cationic, anionic or neutral).
- **Central atom/ion:** the Lewis acid heart —

  $$Fe^{2+}$$

  here — accepting lone pairs into its vacant orbitals. Usually a d-block metal (this is why Unit IV and Unit V travel together).
- **Ligands:** the Lewis bases donating the pairs. **Monodentate** bite once (

  $$NH_3$$

  , Cl^-, H_2O); **bidentate** bite twice (ethane-1,2-diamine "en", oxalate

  $$C_2O_4^{2-}$$

  ); **polydentate** grip with many teeth ($EDTA^{4-}$, six donors — the "hexadentate claw"); **ambidentate** carry two different biting ends and choose one (

  $$NO_2^-$$

  via N or O;

  $$SCN^-$$

  via S or N).
- **Coordination number (CN):** the number of donor atoms bonded to the metal — $[Fe(CN)_6]^{4-}$ has CN = 6; $[Co(en)_3]^{3+}$ also has CN = 6 because each en counts twice. CN = 4 (tetrahedral/square planar) and CN = 6 (octahedral) run the whole course.
- **Coordination sphere:** everything inside the brackets, written as a unit. **Counter ions** ($4K^+$ here) sit outside and balance the charge.
- **Coordination polyhedron:** the spatial arrangement — octahedron for CN 6, tetrahedron or square plane for CN 4.
- **Oxidation number of the central atom:** the charge the metal would carry if all ligands left with their electrons. For

  $$K_4[Fe(CN)_6]$$

  : the entity is 4-, each CN^- is 1-, so $x + 6(-1) = -4 \rightarrow x = +2$.
- **Homoleptic vs heteroleptic:** one ligand type ($\left[Co(NH_3)_6\right]^{3+}$) versus a mixture ($\left[Co(NH_3)_4Cl_2\right]^{+}$). Ligands with negative charges inside the sphere are **anionic ligands** — their names take an -o ending (chlorido, cyanido, oxalato).

Naming Complexes: The IUPAC Machine

The rules compress to a short checklist. Practise on the classics and the machine never jams.

1. **Name the cation first, anion second** — exactly like ordinary salts (

   $$NaCl$$

   : sodium chloride, never the reverse).
2. **Inside the sphere, name ligands first — in alphabetical order** (ignore multiplying prefixes when alphabetising), then the metal.

3. **Prefixes:** di-, tri-, tetra-, penta-, hexa- count simple ligands. For ligands whose own name already contains one of these (as in ethylenediamine), switch to bis-, tris-, tetrakis-.
4. **Anionic ligands end in -o:** chlorido (Cl^-), cyanido (CN^-), hydroxido, oxalato, sulfato. Neutral ligands keep ordinary names except water = **aqua** and ammonia = **ammine**.
5. **The metal's oxidation state follows in Roman numerals in brackets** — with NO space.
6. **If the complex is the anion, the metal name ends in -ate:** ferrate, cuprate, zincate, argentate (silver), stannate, plumbate.
7. **Ambidentate ligands name their biting atom:**

$$NO_2^-$$

through N = nitrito-κN (older: nitro); through O = nitrito-κO (nitrito); SCN^- gives thiocyanato-κS / isothiocyanato-κN.

Formula	Systematic name
$\left[Co(NH_3)_6\right]Cl_3$	hexaamminecobalt(III) chloride
$\left[Co(NH_3)_5\ Cl\right]Cl_2$	pentaamminechloridocobalt(III) chloride
$K_3[Fe(CN)_6]$	potassium hexacyanidoferrate(III)
$K_2[Zn(OH)_4]$	potassium tetrahydroxidozincate(II)
$\left[Pt(NH_3)_2Cl_2\right]$	diamminedichloridoplatinum(II)
$\left[Co(NH_3)_4(H_2\ O)\ Cl\right]Cl_2$	tetraammineaquachloridocobalt(III) chloride
$[Ni(CO)_4]$	tetracarbonylnickel(0)
$\left[Co(en)_2(NH_3)_2\right]^{3+}$	diamminebis(ethane-1,2-diamine)cobalt(III) ion

- **Reverse direction:** "tris(ethane-1,2-diamine)chromium(III) chloride" → the +3 metal with three neutral en ligands → entity +3, so two counter chlorides: $[Cr(en)_3]Cl_3$. Work the charge algebra backwards and the formula writes itself.

Isomerism: Same Formula, Different Molecule

Isomers share a formula but differ in arrangement — and coordination compounds manufacture them in astonishing variety. Two families: **structural** (which atom bonds to which) and **stereoisomerism** (same bonds, different 3-D layout).

Structural isomerism — four varieties

- **Ionisation isomerism:** the counter ion and a sphere ligand swap places, so different ions appear in solution. Violet $\left[Co(NH_3)_5\ Br\right]SO_4$ gives

$$BaSO_4$$

precipitate; red $\left[Co(NH_3)_5SO_4\right]$ Br gives AgBr instead. Same formula, different chemistry.

- **Hydrate (solvate) isomerism:** water moves in and out of the sphere — the three forms of $CrCl_3 \cdot 6H_2O$: violet $[Cr(H_2O)_6]Cl_3$, blue-green $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$, green $[Cr(H_2O)_4Cl_2]Cl \cdot 2H_2O$.
- **Linkage isomerism:** ambidentate ligands switch biting ends — $[Co(NH_3)_5(NO_2)]^{2+}$ (nitro, through N) versus $[Co(NH_3)_5(ONO)]^{2+}$ (nitrito, through O). Red and yellow pair.
- **Coordination isomerism:** needs **both** spheres ionic — the ligands exchange partners between cation and anion: $[Co(NH_3)_6][Cr(CN)_6]$ vs $[Cr(NH_3)_6][Co(CN)_6]$.

Stereoisomerism — geometry and mirrors

- **Geometric (cis/trans):** identical ligands sit adjacent (cis) or opposite (trans).
  - **Square planar** $[Ma_2b_2]$: cis-platin vs trans-platin both exist — the **tetrahedral** $[Ma_2b_2]$ shows **no** cis/trans (all four corners are adjacent). A beloved trap.
  - **Octahedral** $[Ma_4b_2]$: cis and trans (e.g. $[Co(NH_3)_4Cl_2]^+$); $\mathbf{[Ma_3b_3]}$ adds a facial/meridional (fac/mer) pair — three b's on one face vs in a meridian.
- **Optical isomerism:** mirror images that cannot be superimposed — the molecule is chiral. The stars of the syllabus: $\mathbf{[M(en)_3]^{3+}}$ (three bidentates wind into a propeller), **cis-**$\mathbf{[M(en)_2a_2]}$, and octahedral cis-$[M(en)_2b_2]$ — each gives a d/l pair; the **trans** counterpart is achiral (its mirror superimposes). Optical isomers rotate plane-polarised light in opposite directions.

Valence Bond Theory: Hybridising the Metal

VBT asks a simple question: which orbitals of the metal accept the ligand lone pairs? Hybridise them, and the geometry and magnetism follow.

- **The recipe:** count the metal ion's d electrons; see how many d orbitals stay occupied; empty inner $(n-1)d$ orbitals if strong-field ligands can force pairing; hybridise to fit the coordination number — d^2sp^3 (inner orbital, octahedral) or sp^3d^2 (outer orbital, octahedral); sp^3 (tetrahedral) or dsp^2 (square planar).
- **Inner orbital example — $\mathbf{[Co(NH_3)_6]^{3+}}$:**

  $$Co^{3+}$$

  is $3d^6$. Ammonia is a strong-field ligand: the four unpaired d electrons pair up, leaving two empty 3d orbitals. Hybridise $3d^2 + 4s + 4p^3 = d^2sp^3$ — octahedral, and magnetism shows **zero** unpaired electrons (diamagnetic). "Inner" because inner d orbitals did the bonding.
- **Outer orbital example — $\mathbf{[CoF_6]^{3-}}$:** F^- is weak-field — no pairing happens. The 3d set stays half-occupied (4 unpaired), so the metal must use the **outer** 4d orbitals: sp^3d^2 octahedral, strongly paramagnetic.
- **The magnetic test closes the loop:** measure the moment, count the unpaired electrons, and you have identified inner vs outer orbital — VBT's one genuinely useful trick.
- **Limitations (the syllabus lists them):** it says **nothing** about colour; it gives no ranking of ligands (why CN^- pairs but F^- does not); it ignores the elegant geometry of every ligand's own orbitals; and it cannot predict how many isomers a complex really has. Crystal field theory steps into that gap.

Crystal Field Theory: Splitting the d Orbitals

CFT's move is brutal and effective: forget shared electrons — treat every ligand as a **point charge**. In an octahedral complex, six negative points arrive along the x, y and z axes. The five d orbitals respond differently: the two pointing **at** the ligands (e_g: $d_{x^2-y^2}$, d_{z^2}) feel repulsion and rise; the three sitting **between** them (t_2 g: d_{xy}, d_{yz}, d_{zx}) dodge and sink. The energy gap is the **crystal field splitting**, Δ_o.

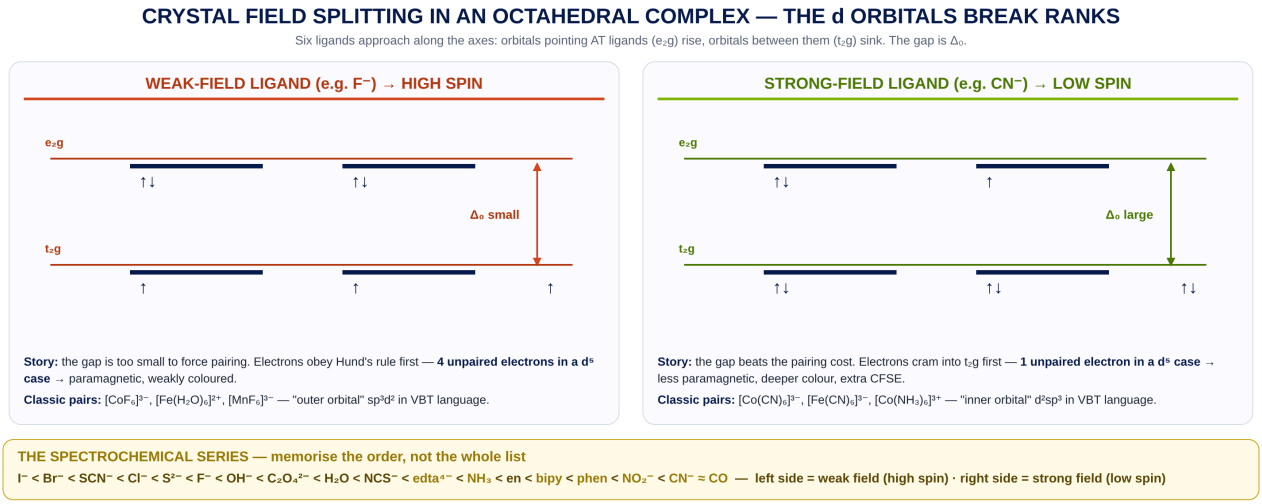


Figure 1: Octahedral splitting: weak-field ligands leave a small Δ_o and a high-spin d^5 fill; strong-field ligands widen the gap, pair the electrons up, and produce a low-spin fill. The spectrochemical series ranks the ligands.

- **Fill the orbitals and count the story:** electrons enter t_2 g first; whether the fourth electron pairs in t_2 g or climbs to e_g depends on the duel between pairing energy P and Δ_o.
  - $\Delta_o < P$ **(weak field):** spread out — **high spin**, more unpaired electrons, more paramagnetic.
  - $\Delta_o > P$ **(strong field):** pair up — **low spin**, fewer unpaired electrons.
- **The spectrochemical series** ranks ligands by the gap they create:

$$\mathrm{Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2\,O < NCS^- < edta^{4-} < NH_3 < en < bipy < phen < NO_2^- < C}$$

- **Colour, explained at last:** an electron absorbs a photon and jumps t_2 g $\rightarrow e_g$; the absorbed frequency obeys

$$h\nu = \Delta_o$$

. Bigger gap, higher-energy (bluer) light absorbed, so the complex reflects the complementary colour. Changing ligands changes Δ_o — $\left[\mathrm{Ti(H_2\,O)_6}\right]^{3+}$ is violet, and swapping in stronger-field ligands shifts its colour systematically.
- **Magnetism, predicted:** count unpaired electrons after filling (5 unpaired for high-spin d^5 1 for low-spin d^5) and apply the spin-only formula

$$\mu = \sqrt{n(n+2)}$$

from Unit IV.
- **Tetrahedral and square planar:** tetrahedral splitting also produces two sets (e below t_2) but the gap is small — **no low-spin tetrahedral complexes exist** (another favourite check). Square planar splitting is the deepest of all — favoured by d^8 ions like

$$Pt^{2+}$$

,

$$Ni^{2+}$$

.

- **Limitations:** CFT ignores covalent character entirely (ligands are not really point charges), cannot explain why neutral CO beats anionic F^- so badly in the series, and stays silent on π-bonding. Its predictions on colour and magnetism, however, are superb.

Metal Carbonyls: The Synergic Handshake

Metal carbonyls —

$$Ni(CO)_4$$

,

$$Fe(CO)_5$$

— are homoleptic complexes of neutral CO with the metal in oxidation state zero, and their bonding is the chapter's most elegant idea.

- **Shapes:**

  $$Ni(CO)_4$$

  is **tetrahedral** (sp^3-hybridised Ni(0), $3d^{10}$ — diamagnetic);

  $$Fe(CO)_5$$

  is **trigonal bipyramidal** (dsp^3). All metal carbonyls are famously volatile liquids or solids — a hint of covalent, non-ionic character.
- **The two-way bond (synergic bonding):** CO donates its carbon lone pair to the metal (σ bond: metal ← C) **and** the metal donates a pair of its d electrons into CO's empty antibonding π orbital (π back-donation: metal → C). Each flow strengthens the other — a feedback handshake — which is why the M–C bond ends up **shorter and stronger** than a plain single bond, while CO's own C–O bond weakens slightly (measurable by infrared).
- **Why zero oxidation states are stable:** the back-donation spreads electron density that would otherwise pile up on the metal — CO is both a σ donor and a π acceptor, and that dual role stabilises metals that normally refuse complexing.

Why Coordination Chemistry Runs the World

- **Extraction of silver and gold:** the Mac-Arthur–Forrest cyanide process — crushed ore + dilute NaCN + air dissolves the metal as $[Au(CN)_2]^-$; zinc dust then displaces the gold. A complex formation is the whole mining trick:

$$Au(s) + 2CN^-(aq) + \frac{1}{2}O_2(g) + H_2\ O \rightarrow [Au(CN)_2]^- + 2OH^-$$

- **Medicine: cis-platin** ($\left[Pt(NH_3)_2Cl_2\right]$, the cis isomer!) is a frontline anti-cancer drug — it cross-links DNA strands of dividing cells (the trans isomer is ineffective — cis/trans decides

therapy). **EDTA** is the antidote for lead poisoning: one hexadentate claw wraps the metal and the kidneys flush the complex out (chelation therapy).

- **Biological metal centres:** haemoglobin's Fe^{2+} grabs

$$O_2$$

(CO's tighter binding is why carbon monoxide poisons); chlorophyll's Mg^{2+} harvests sunlight; vitamin B^{12}'s Co centre carries out methyl transfers. Life is coordination chemistry wearing a lab coat.

- **Analytical chemistry:**

$$Ni^{2+}$$

confirmed by bright-red precipitate with dimethylglyoxime (DMG);

$$Fe^{3+}$$

by blood-red thiocyanato complex; water hardness titrated with EDTA; $\left[Ag(NH_3)_2\right]^+$ in Tollen's reagent distinguishing aldehydes. Every spot test is a complex forming.

- **Purification:** nickel refined by the Mond process through volatile

$$Ni(CO)_4$$

— the metal travels as a coordination compound and deposits pure at 200°C.

Key Reactions & Rules: One-Stop Revision

DEFINITIONS, NAMES, SPIN PREDICTIONS

Rule / formula	Symbols in plain words	When to reach for it
Oxidation number = charge algebra	entity charge = metal + ligand charges	Every naming and formula task
CN counts donor atoms	en and oxalate count TWICE; EDTA six times	$[Co(en)_3]^{3+}$ has CN = 6, not 3
Ligand alphabet, di/tri/tetra	ignore prefixes when alphabetising	All IUPAC names
Anionic metal: -ate	ferrate, cuprate, zincate, argentate	Complex anions like $K_4[Fe(CN)_6]$
aqua = H_2O; ammine = NH_3	neutral ligand special names	Naming neutral-ligand complexes
Ionisation isomer	counter ion swaps with sphere ligand	$Co(NH_3)_5BrSO_4$ type pairs
Linkage isomer	ambidentate flips bite end	nitro vs nitrito; thiocyanato vs isothiocyanato

Rule / formula	Symbols in plain words	When to reach for it
Square planar Ma^2b^2: cis/trans	tetrahedral Ma^2b^2: NONE	The cis/trans existence check
Octahedral Ma^4b^2: cis/trans; Ma^3b^3: fac/mer	adjacent vs opposite; one face vs meridian	Geometric isomer counting
Optical: $[M(\text{en})_3]^{3+}$, cis-$M(\text{en})_2a_2$	propeller chirality; trans is achiral	Mirror-image questions
VBT: d^2sp^3 inner vs sp^3d^2 outer	strong field pairs up (low spin); weak field uses outer d	Hybridisation from magnetism
Δ_o vs P decides spin	small gap: high spin; big gap: low spin	CFT electron filling
Spectrochemical series	I^- weakest … CN^- ≈ CO strongest	Ranking ligand strength
Colour: $h\nu = \Delta_o$	absorbed light matches the gap	Colour-shift predictions
$\mu = \sqrt{n(n+2)}$ BM	count unpaired electrons after filling	Magnetic moment of any complex
Synergic bond in M–CO	σ donation + π back-donation	Carbonyl bond-strength questions

Exceptions & Traps

THE TRAPS THIS CHAPTER SETS — READ BEFORE THE MOCK

The rule you assume	The exception / twist	The trap in the question
Oxidation number = coordination number	Independent numbers — Co^{3+} with CN 6 in most ammines, but Cu^{2+} with CN 4	Compute charges first, count donors second — never confuse the two
Ligand prefixes are alphabetised	Alphabetise by the NAME, ignoring di/tri/tetra	"tetraammineaquachlorido…" is a-, then c-, then… check the rule
Tetrahedral Ma^2b^2 shows cis/trans	Only square planar does — all tetrahedral corners are adjacent	cis-platin exists; tetrahedral "cis" does not

The rule you assume	The exception / twist	The trap in the question
All ambidentate ligands isomerise freely	Only NO_2^-, SCN^- (CN^- rarely) show linkage isomerism	NH_3 or Cl^- options offered as "linkage" answers are bait
Strong-field ligands are anions	CO and NH_3 outrank every anion	The spectrochemical series' top entries are NEUTRAL molecules
Complex = double salt	Double salts lose identity in solution; complexes keep it	The AgCl test: $K_4[Fe(CN)_6]$ gives no Fe^{2+} precipitate with the usual reagents
Low spin needs a metal in a high oxidation state only	The LIGAND decides via Δ_o vs P	Fe^{3+} is high-spin with F^- but low-spin with CN^-
$Ni(CO)_4$ is paramagnetic like most complexes	Ni(0) is $3d^{10}\ 4s^0$ — diamagnetic tetrahedral	"zero unpaired electrons" answers for carbonyls

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The Whole Chapter in One Picture

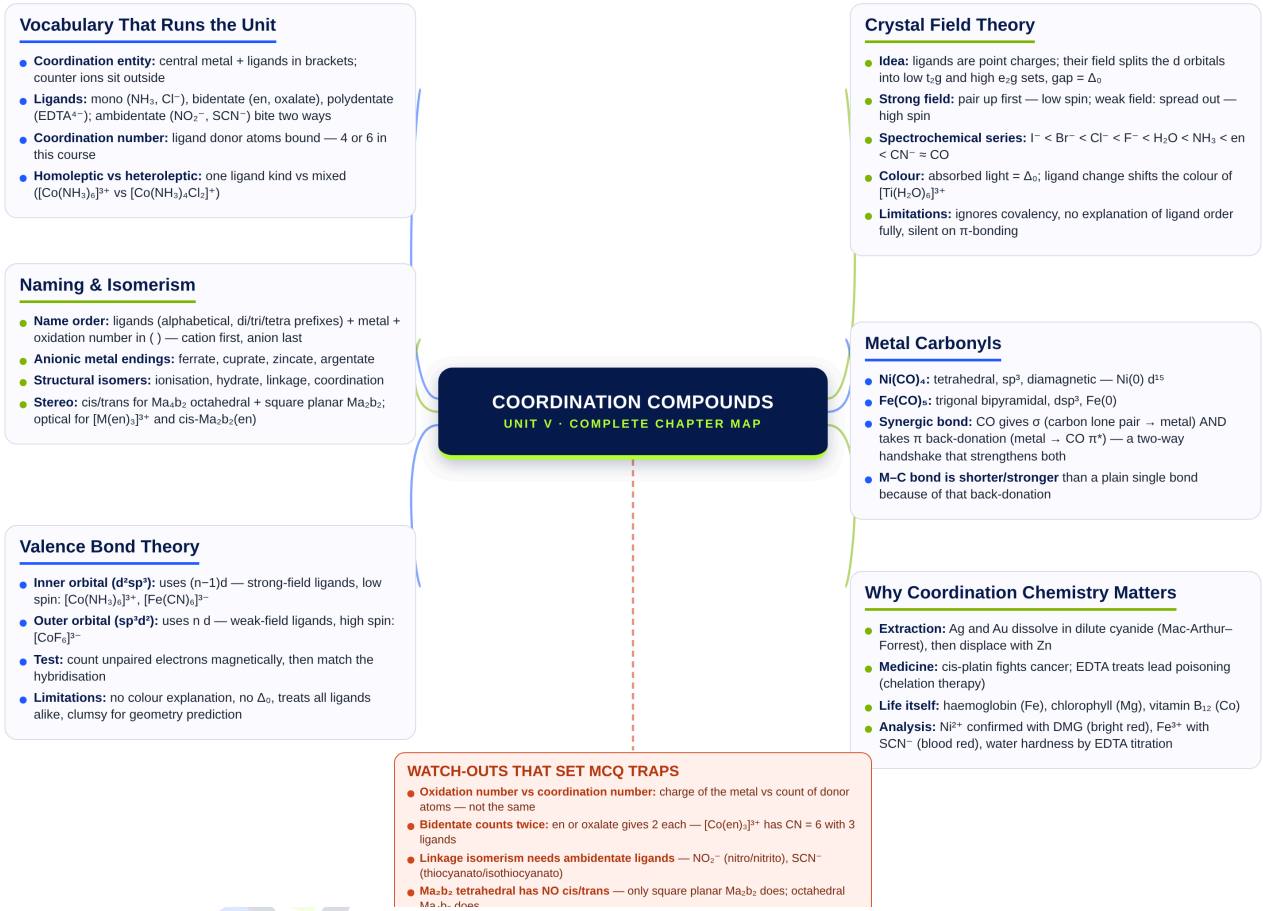


Figure 2: Chapter mindmap: vocabulary, naming and isomerism, VBT inner/outer orbitals, CFT splitting and the spectrochemical series, metal carbonyls, and applications — with the traps that catch most students.

Practice Questions (Exam Pattern)

Ten questions in the CUET UG style. Attempt all of them — the paper gives +5 for each correct answer and −1 for every incorrect answer.

Q1. The coordination number and oxidation state of cobalt in $[Co(en)_2Cl_2]\,Cl$ are respectively:

(A) 4 and +3
(B) 6 and +3
(C) 6 and +2
(D) 4 and +2

Q2. Which of the following behaves as a double salt rather than a complex in aqueous solution?

(A) $K_4[Fe(CN)_6]$
(B) $KCl \cdot MgCl_2 \cdot 6H_2O$
(C) $[Cu(NH_3)_4]SO_4$
(D) $[Co(NH_3)_6]Cl_3$

Q3. The IUPAC name of $K_3[Fe(CN)_6]$ is:

(A) potassium hexacyanidoiron(III) **(B)** potassium hexacyanidoferrate(III)
(C) tripotassium hexacyanidoironate(II) **(D)** potassium ferricyanide(III)

Q4. The pair $\left[Co(NH_3)_5(NO_2)\right]^{2+}$ and $\left[Co(NH_3)_5(ONO)\right]^{2+}$ exemplifies:

(A) ionisation isomerism **(B)** linkage isomerism
(C) coordination isomerism **(D)** hydrate isomerism

Q5. A tetrahedral complex of the type [Mabcd] with four different ligands:

(A) shows cis/trans isomerism **(B)** shows optical isomerism
(C) shows both **(D)** shows neither

Q6. In $[CoF_6]^{3-}$ the cobalt(III) ion ($3d^6$) retains four unpaired electrons. The hybridisation of Co is therefore:

(A) d^2sp^3 (inner orbital) **(B)** sp^3d^2 (outer orbital)
(C) sp^3 **(D)** dsp^2

Q7. According to the spectrochemical series, the STRONGEST field ligand among these is:

(A) F^- **(B)** H_2O
(C) NH_3 **(D)** CN^-

Q8. The colour of a complex arises because:

(A) electrons jump between the split d-orbital sets by absorbing visible light **(B)** ligands emit coloured light
(C) the metal ion reflects its pure colour **(D)** solvent molecules vibrate

Q9. The bonding in $Ni(CO)_4$ involves:

(A) only σ donation from CO **(B)** only π back-donation from Ni
(C) synergic σ donation and π back-donation **(D)** purely ionic attraction

Q10. cis-platin, the anti-cancer drug, is correctly written as:

(A) trans-$\left[Pt(NH_3)_2Cl_2\right]$ **(B)** cis-$\left[Pt(NH_3)_2Cl_2\right]$
(C) $[Pt(CN)_4]^{2-}$ **(D)** $K_2[PtCl_6]$

Detailed Solutions

Sol. Q1. Two en ligands bite twice ($2 \times 2 = 4$ donor atoms) plus two Cl^- = CN 6. Charges: entity +1 (one counter Cl^-), each Cl^- is −1, so Co = +1 + 1 + 1 = +3. CN 6, oxidation state +3.

Answer: (B)

Sol. Q2. Carnallite falls apart in water into K^+, Mg^{2+} and Cl^- — every ion testable, identity only in the crystal: a double salt. The other three keep their bracketed entities intact in solution.

Answer: (B)

Sol. Q3. The complex is the anion, so the metal takes -ate: ferrate. Six cyanido ligands, Fe in +3 (charge algebra: -3 entity $= x + 6(-1) \rightarrow x = +3$), potassium outside first: potassium hexacyanidoferrate(III).

Answer: (B)

Sol. Q4. The ambidentate NO_2^- has flipped its biting atom — N-bonded nitro vs O-bonded nitrito — with identical formula: linkage isomerism. Ionisation isomers would swap the counter ion; there is none here (the entity is its own cation).

Answer: (B)

Sol. Q5. Four different ligands on a tetrahedron: every arrangement is a mirror image that cannot be superimposed — optical isomers. No cis/trans (all corners adjacent); that trick belongs to square planar Ma^2b^2.

Answer: (B)

Sol. Q6. Four unpaired 3d electrons means pairing never happened — F^- is weak-field. Bonding must therefore use outer 4d orbitals: sp^3d^2, an outer-orbital (high-spin) octahedral complex.

Answer: (B)

Sol. Q7. The series ends … en $<$ bipy $<$ phen $<$ NO_2^- $<$ CN^- $\approx$ CO. Cyanide tops the list — it produces the largest Δ_o, forcing low-spin configurations.

Answer: (D)

Sol. Q8. CFT's colour story: a d electron absorbs a photon of energy exactly Δ_o and jumps $t_2g \rightarrow e_g$. The transmitted light is the complement — change the ligand, change Δ_o, change the colour.

Answer: (A)

Sol. Q9. CO donates its carbon lone pair to Ni (σ) AND Ni back-donates d-electron density into CO's empty π^* orbital (π back-donation). Each flow strengthens the other — the synergic bond — making the M–C bond unusually short and strong.

Answer: (C)

Sol. Q10. Only the cis isomer positions the two chlorides close enough to cross-link DNA — trans-platin is therapeutically useless. The drug is cis-diamminedichloridoplatinum(II).

Answer: (B)

Wrong-Answer Clinic — the one-line corrections

Dropped marks above? Find your slip here and rewrite the corrected rule in your own words.

If you answered...	Your slip was...	The corrected rule, in one line
Q1 option A	Counting en as one donor	Bidentate = two donor atoms each: en × 2 gives CN 4 from just two ligands
Q2 option A	Trusting the bracketed look	The solution test decides: double salts release all ions; complexes keep the bracketed unit alive
Q4 option A	Spotting two "parts" and guessing ionisation	Linkage needs an ambidentate ligand flipping its bite — NO_2^- is the chapter's star
Q6 option A	Assuming octahedral always means inner	High spin (unpaired electrons left) forces OUTER sp^3d^2 — pairing is what unlocks inner d orbitals
Q8 option B	Blaming the ligands' own colour	Free ligands barely absorb visible light; the d–d jump inside the split subshell is the whole story

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